

NETHOLABS

Functional Brain Emulation

Data Overview

Multi-Modal Neural Recording Stack — Technical Reference

March 2026

67 GB TOTAL DATA	42 XDF FILES	9 SUBJECTS	11 MODALITIES	500+ CHANNELS	10 DEVICES
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RECORDING DEVICES

DEVICE	SIGNAL TYPE	CHANNELS	SAMPLE RATE	INTERFACE	NOTES
Kernel Flow 2	TD-fNIRS	48 modules × 3 streams	5 Hz	WebSocket (Kortex SDK)	690nm (Red) + 850nm (IR) + Quality. ws://127.0.0.1:13254
DSI-24	EEG (dry electrode)	24 ch	300 Hz	Bluetooth (COM3)	Wearable Sensing. Full 10-20 montage. Mobile-ready.
Gazepoint GP3	Eye tracking (desktop)	6 ch	60 Hz	TCP 127.0.0.1:4242	Fixation XY, pupil diameter, best POG, validity. XML protocol.
Tobii Pro Glasses 3	Eye tracking + IMU (mobile)	10 + 9 ch	50 + 100 Hz	WiFi WebSocket	Binocular gaze, pupil. Accel/gyro/magnetometer.
Pupil Labs Neon	Eye tracking + IMU (mobile)	17 + 10 ch	200 + 110 Hz	WiFi RT API	Gaze, blink, fixation, 3D eye. Scene camera + iris video.
Delsys Trigno	Surface EMG + Aux	16 + 48 ch	2000 + 148.1 Hz	TCP SDK	Wireless sensors. EMG + tri-axial accelerometer per sensor.
StretchSense	Hand & body kinematics	224 ch total	Variable	BLE	56ch left hand + 56ch right hand + 112ch body suit.
Microphone	Audio	1 ch	44100 Hz	USB	System microphone. Voice + ambient audio.

Device	Signal Type	Channels	Sample Rate	Interface	Notes
USB Webcam	Video (participant face)	640×480	30 fps	USB	Frontal face view. Saved as AVI alongside XDF.
Raspberry Pi 5 (×2)	Video (scene cameras)	2 ch each	30 fps	WiFi (MJPEG + LSL)	Pi Camera Module 3. Frame timestamps via LSL.

LSL Stream Inventory

LSL Stream Name	Source	Channels	Rate	Content	Bridge Script
 DSI24	DSI-24	24	300 Hz	EEG (10-20 montage, all electrodes)	dsi24_lsl.py
 KernelFlow_Red	Kernel Flow 2	48	5 Hz	HbO oxygenation (690nm wavelength)	kernel_flow_lsl.py
 KernelFlow_IR	Kernel Flow 2	48	5 Hz	HbR deoxygenation (850nm wavelength)	kernel_flow_lsl.py
 KernelFlow_Quality	Kernel Flow 2	48	5 Hz	Per-module signal quality index	kernel_flow_lsl.py
 Gazepoint	GP3	6	60 Hz	FPOG X/Y, FPOGD, FPOGV, BPOG X/Y	gazepoint_lsl.py
 TobiiGlasses_Gaze	Tobii Glasses 3	10	50 Hz	Binocular gaze position + pupil diameter	tobii_glasses_lsl.py
 TobiiGlasses_IMU	Tobii Glasses 3	9	100 Hz	Accelerometer + gyroscope + magnetometer	tobii_glasses_lsl.py
 PupilNeon_Gaze	Pupil Neon	17	200 Hz	Gaze, blink, fixation, 3D eye position	pupil_neon_lsl.py
 PupilNeon_IMU	Pupil Neon	10	110 Hz	Accelerometer + gyroscope + quaternion	pupil_neon_lsl.py
 Delsys_EMG	Delsys Trigno	16	2000 Hz	Surface electromyography	delsys_emg_lsl.py
 Delsys_AUX	Delsys Trigno	48	148.1 Hz	Accelerometer + auxiliary analog	delsys_emg_lsl.py
 StretchSense	StretchSense	224	Var.	Left hand (56) + right hand (56) + body (112)	(native LSL)
 Microphone	System mic	1	44100 Hz	Audio waveform	(native LSL)

LSL Stream Name	Source	Channels	Rate	Content	Bridge Script
● StimulusMarkers	PsychoPy	Markers	Event	Stimulus onset, offset, response events	stimulus_system/
● PiCamera / PiCamera2	RPi 5	2 each	30 Hz	Frame timestamps (video stored on Pi)	pi_camera_lsl.py

DATA PIPELINE

SENSORS¹

10 hardware devices generate raw neural, ocular, muscular, kinematic, and audio/video signals via proprietary SDKs and protocols.

BRIDGE SCRIPTS²

Python scripts (*_lsl.py) translate each device SDK into LSL outlets. Running on Windows 11 with Python 3.11. Located in C:\Users\chris\LSL_Tools\.

LSL TRANSPORT³

Lab Streaming Layer unifies all streams with sub-millisecond network clock synchronization. pylsl 1.18.1. Up to 15 concurrent streams.

RECORDING⁴

LabRecorder captures all active LSL streams to .xdf files with BIDS-compliant naming. Stored in Documents\CurrentStudy\.

VISUALIZATION⁵

Real-time PyQt5 dashboards (lab + mobile). Offline Three.js 3D demo. Cinematic video exports via OpenCV.

MODALITY SUMMARY

EEG

24 channels
300 Hz
DSI-24 (dry)

fNIRS

144 channels (3x48)
5 Hz
Kernel Flow 2

Eye Tracking

6-17 channels
50-200 Hz
GP3 / Tobii / Neon

EMG

16 + 48 aux channels
2000 Hz
Delsys Trigno

Hand Kinematics

112 channels (L+R)
Variable rate
StretchSense gloves

Body Motion

112 channels
Variable rate
StretchSense suit

IMU

9-10 channels
100-110 Hz
Tobii / Neon

Audio & Video

1 audio + 5 video
44.1kHz / 30fps
Mic + cameras

RECORDING SESSIONS

SUBJECT	FILES	SIZE	STREAMS PRESENT	NOTES
session_* (lab)	14	3.1 GB	KF_Red, KF_IR, KF_QA, Gazepoint, StretchSense, Mic	Feb 27 – Mar 6, 2026. 8 modalities per session.
sub-P001	6	13.8 GB	Full lab stack	Includes 5 _old versions. Largest single subject.
sub-P002	2	3.7 GB	Full lab stack	Includes 1 _old version.
sub-P007	1	1.7 GB	Full lab stack	Single clean session.
sub-090	3	1.5 GB	Varies per session	3 sessions (longitudinal Baseline task).
sub-09031	1	732 MB	Baseline task	Single session.
sub-0999	2	176 MB	Baseline task	Includes 1 _old version.
sub-2003	3	46 MB	Baseline task	2 sessions, repeat participant.
sub-P009	1	56 KB	Minimal	Likely aborted recording.